

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 31-08-2026

1. TITLE

Design of a Basic Webpage using HTML Tags

2. AIM

To develop a webpage displaying college information utilizing fundamental HTML tags such as headings, paragraphs, links, lists, tables, images, forms, and multimedia elements.

3. OBJECTIVE

The objectives of this experiment are:

- To understand the basic structure of an HTML document.
- To implement text formatting, hyperlinks, and multimedia tags.
- To organize tabular data using HTML tables.
- To build a user input form for student registration.

4. THEORY

HTML (HyperText Markup Language) is the standard markup language for creating web pages. It provides structural semantics for text and media. Elements like <header>, <nav>, <section>, <article>, and <footer> offer semantic meaning, while tags like <table>, , <audio>, and <video> enable embedding of data and media. Forms allow collection of user input via various input types.

5. ALGORITHM / PROCEDURE

1. Create index.html and define the basic document structure.
2. Add a header and navigation links to different sections.
3. Use text formatting tags (bold, italic, sub, sup) in the about section.
4. Implement ordered and unordered lists for courses and facilities.
5. Add internal and external hyperlinks using the anchor tag.
6. Design a student details table with borders and cell padding.
7. Embed images for the college logo and campus.
8. Create a registration form with various input types.
9. Add audio and video controls for multimedia playback.
10. Test and verify the page structure in a browser.

6. CODE EXPLANATION

The HTML document starts with the `<!DOCTYPE html>` declaration, specifying it as an HTML5 document. The `<a>` (anchor) tags are utilized to create hyperlinks for navigation across sections. A `<table>` element is implemented to present tabular data with clearly defined headers and rows. The `<form>` section contains various interactive input fields such as text, email, date, radio buttons, and select dropdowns to collect user data effectively. Lastly, multimedia elements like `<video>` and `<audio>` are embedded using their respective tags to enable media playback directly within the webpage.

7. PROGRAM

index.html

```

<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>

<body>

  <!-- ===== HEADER ===== -->
  <header>
    <h1>ANITS - College Information Portal</h1>

    <p>
      Welcome to <b>Anil Neerukonda Institute of Technology and Sciences (ANITS)</b>,
      Visakhapatnam.
    </p>
  </header>

  <!-- ===== NAVIGATION ===== -->
  <nav>
    <a href="#about">About</a> |
    <a href="#courses">Courses</a> |
    <a href="#students">Students</a> |
    <a href="#register">Registration</a>
  </nav>

  <hr>

  <!-- ===== SECTION ===== -->
  <section id="about">

    <article>

      <h2>About ANITS</h2>

      <p>
        <b>ANITS</b> is one of the premier engineering colleges in Andhra Pradesh.
        It offers quality education with excellent laboratories,
        experienced faculty and placement opportunities.
      </p>

      <h3>Text Formatting Example</h3>

      <p>
        <b>Vinay</b> (boldtext) <br>
        <i>Vijay</i> (Italic Text) <br>
        <u>jyothi</u> (Underlined Text) <br>
        H<sub>2</sub>O (Subscript) <br>
        x<sup>2</sup> + y<sup>2</sup> (Superscript)
      </p>

    </article>

  </section>

```

```

<hr>

<!-- ===== LISTS ===== -->

<section id="courses">

    <h2>Courses Offered (Ordered List)</h2>

    <ol>
        <li>CSE</li>
        <li>CSE (AI&ML)</li>
        <li>CSD</li>
        <li>MECH</li>
        <li>EEE</li>
    </ol>

    <h2>College Facilities (Unordered List)</h2>

    <ul>
        <li>Central Library</li>
        <li>Computer Labs</li>
        <li>Hostel</li>
        <li>play ground</li>
        <li>Transport Facility</li>
    </ul>

</section>

<hr>

<!-- ===== HYPERLINKS ===== -->

<h2>Useful Links</h2>

<a href="https://anits.org/" target="_self">
    Visit ANITS Website
</a>

<br><br>

<a href="https://www.google.com" target="_blank">
    Visit Google
</a>

<br><br>

<a href="https://www.youtube.com" target="_parent">
    Visit youtube
</a>
<hr>

<!-- ===== TABLE ===== -->

<section id="students">

    <h2>Student Details</h2>

    <table border="1" cellpadding="10">

```

```

        <tr>
            <th>Roll No</th>
            <th>Name</th>
            <th>Branch</th>
            <th>Email</th>
        </tr>

        <tr>
            <td>101</td>
            <td>Vinay</td>
            <td>CSE (AI&ML)</td>
            <td>vinay@gmail.com</td>
        </tr>

        <tr>
            <td>102</td>
            <td>Santhosh</td>
            <td>CSE</td>
            <td>santhosh@gmail.com</td>
        </tr>

        <tr>
            <td>103</td>
            <td>Vijay</td>
            <td>MECH</td>
            <td>vijay@gmail.com</td>
        </tr>

    </table>

</section>

<hr>

<!-- ===== IMAGES ===== -->

<h2>College Logo</h2>



<br><br>

<h2>College Campus</h2>



<hr>

<!-- ===== FORM ===== -->

<section id="register">

    <h2>Student Registration Form</h2>

```

```
<form>

    <label>Name:</label><br>
    <input type="text" placeholder="Enter your name">
    <br><br>

    <label>Email:</label><br>
    <input type="email" required placeholder="Enter your email">

    <br><br>

    <label>Password:</label><br>
    <input type="password">

    <br><br>

    <label>Date of Birth:</label><br>
    <input type="date">

    <br><br>

    <label>Age:</label><br>
    <input type="number">

    <br><br>

    <label>Favourite Color:</label><br>
    <input type="color">

    <br><br>

    <label>Programming Skill:</label><br>
    <input type="range">

    <br><br>

    <label>Gender:</label><br>

    <input type="radio" name="gender"> Male

    <input type="radio" name="gender"> Female

    <input type="radio" name="gender"> Other

    <br><br>

    <label>Hobbies:</label><br>

    <input type="checkbox"> Reading

    <input type="checkbox"> Coding

    <input type="checkbox"> Sports

    <br><br>

    <label>Branch:</label>
```

```

        <select>

            <option>CSE</option>

            <option selected>CSE (AI&ML)</option>

            <option>ECE</option>

            <option>EEE</option>

            <option>IT</option>

        </select>

        <br><br>

        <label>Comments:</label><br>

        <textarea rows="5" cols="40"></textarea>

        <br><br>

        <input type="submit" value="Submit">

        <input type="reset" value="Reset">

    </form>

</section>

<hr>

<!-- ===== AUDIO ===== -->

<h2>College Audio</h2>

<audio controls>

    <source src="college.mp3" type="audio/mpeg">

</audio>

<hr>

<!-- ===== VIDEO ===== -->

<h2>College Video</h2>

<video width="500" controls>

    <source src="https://youtu.be/LzKjue8PchA" type="video/mp4">

</video>

<hr>

<!-- ===== FOOTER ===== -->

```

```

<footer>

    <p>
        © 2026 ANITS | College Information Portal
    </p>

</footer>

</body>

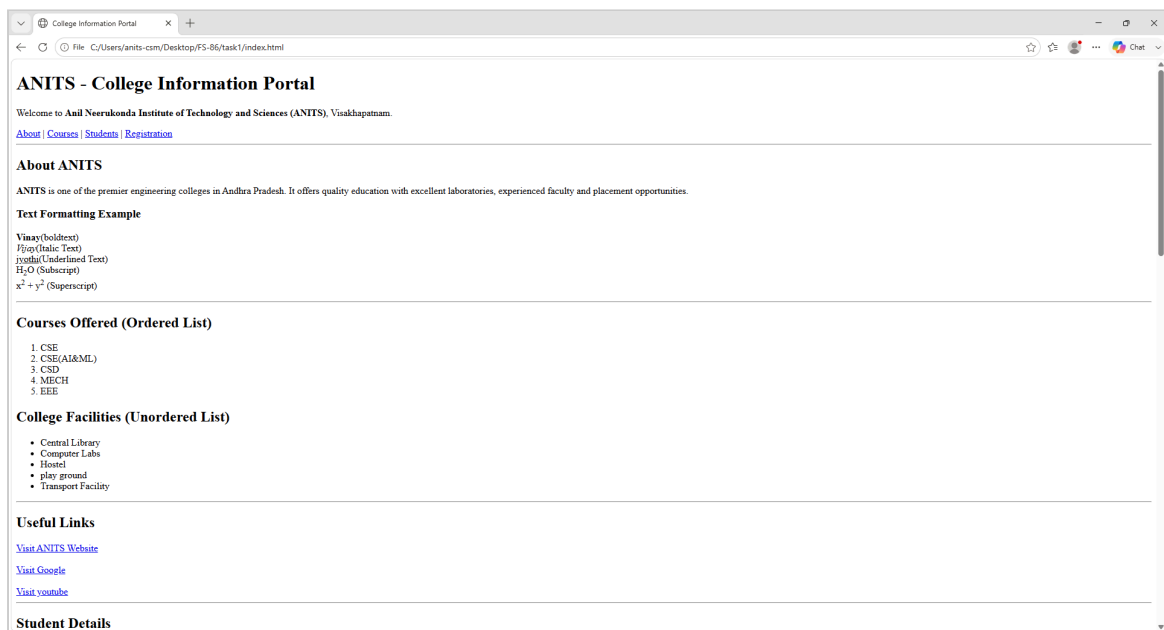
</html>

```

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Form Elements	Verify all input fields	Form fields render correctly	Pass
TC-02	Hyperlinks	Click on links	Links open correct targets	Pass
TC-03	Multimedia	Play audio/video	Media plays successfully	Pass

9. OUTPUT



College Information Portal

File C:/Users/anits-csm/Desktop/FS-86/task1/index.html

Student Details

Roll No	Name	Branch	Email
101	Vinay	CSE(AI&ML)	vinay@gmail.com
102	Santhosh	CSE	santhosh@gmail.com
103	Vijay	MECH	vijay@gmail.com

College Logo



College Campus



Student Registration Form

Name:

Email:

Password:

College Information Portal

File C:/Users/anits-csm/Desktop/FS-86/task1/index.html

Date of Birth:

Age:

Favourite Color:

Programming Skill:

Gender:
☐ Male ☐ Female ☐ Other

Hobbies:
☐ Reading ☐ Coding ☐ Sports

Branch:

Comments:

College Audio

0:00 / 0:00

College Video

0:00

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10. RESULT

Thus, a comprehensive HTML webpage incorporating text formatting, lists, tables, forms, and multimedia elements was successfully designed and verified.